

Fraud & Anomaly Detection

Cause-first risk framework using real G'Contest banking data

1,418,030

Transactions scored

182,546

High/Critical cases

25,194

Customers impacted

- No synthetic data and no fake fraud labels.
- Output is a prevention queue with explainable risk reasons.
- Framework aligns with risk-based banking controls: step-up authentication, manual review, AML escalation.

Why Cause-First

BGK note: identify root causes before introducing the model

- Account takeover / identity compromise: new device, new IP, night access, new beneficiary, late-stage digital activity.
- Unauthorized transfer / capital outflow: outside-bank transfer, amount above customer baseline, daily burst, cash-out versus CASA.
- AML network / mule-account pattern: shared IP/device/beneficiary, round high-value transfers, repeated external transfers.

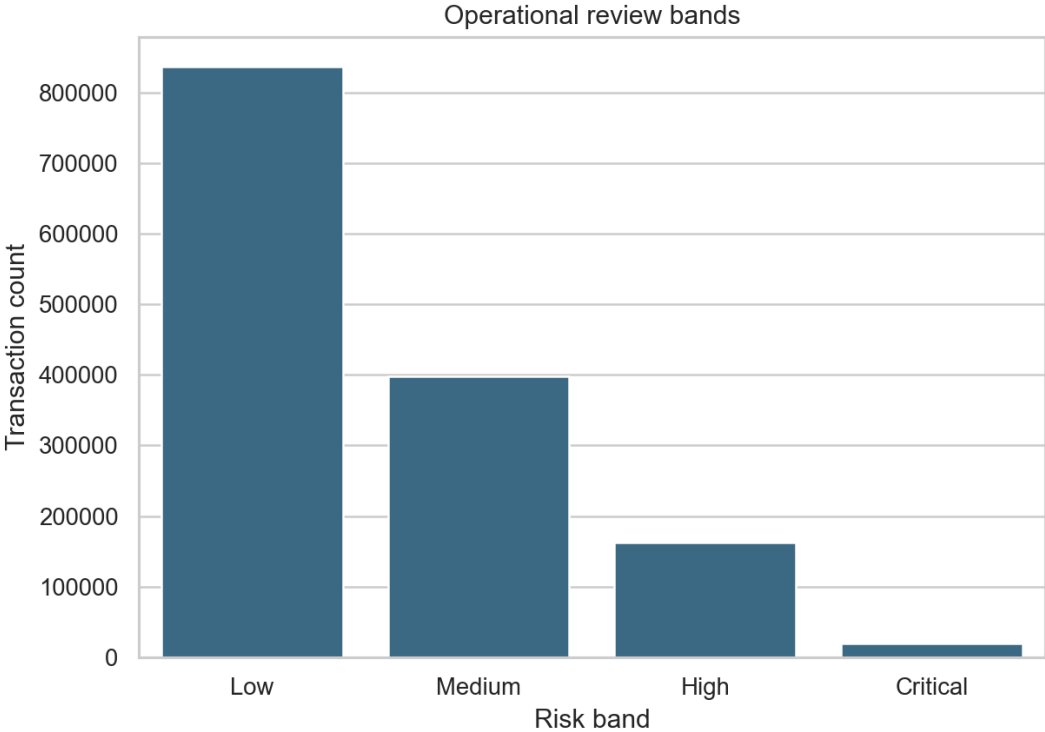
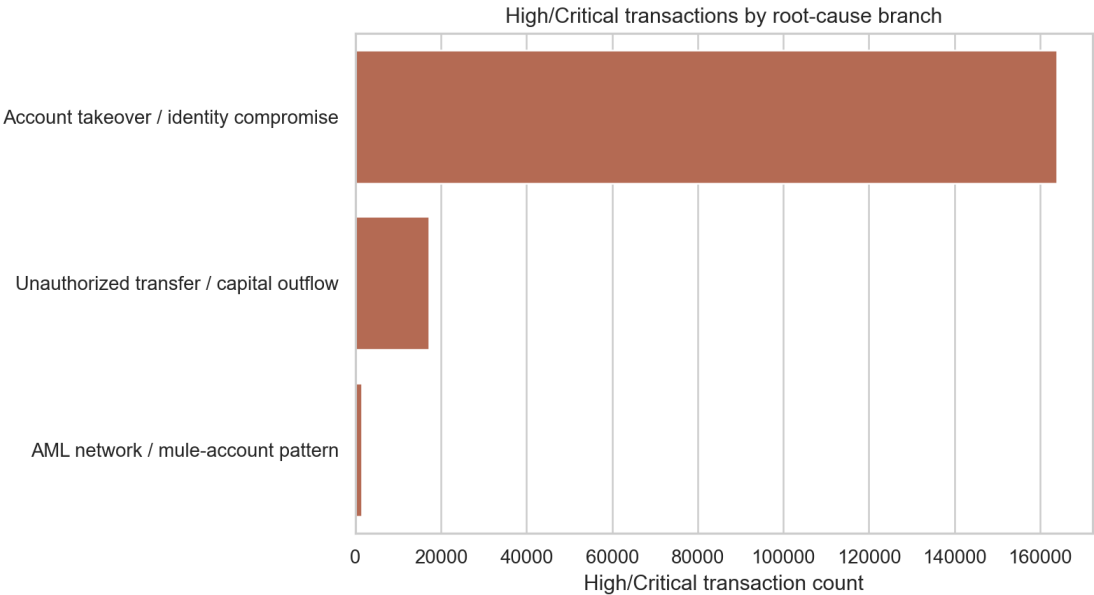
Framework

Data dictionary -> behavioral baseline -> root-cause rules -> weak labels -> prevention model

- | | |
|-----------------|---|
| 1. Data | Customer, Transaction, Activity, Deposit, Lending, Card |
| 2. Baseline | Customer amount P95, daily frequency, known device/IP/beneficiary |
| 3. Root causes | ATO, unauthorized transfer, AML network scores |
| 4. Model | Supervised classifier learns from rule-derived weak labels |
| 5. xAI + Action | Reason codes, SHAP surrogate, Allow/Monitor/Step-up/Block |

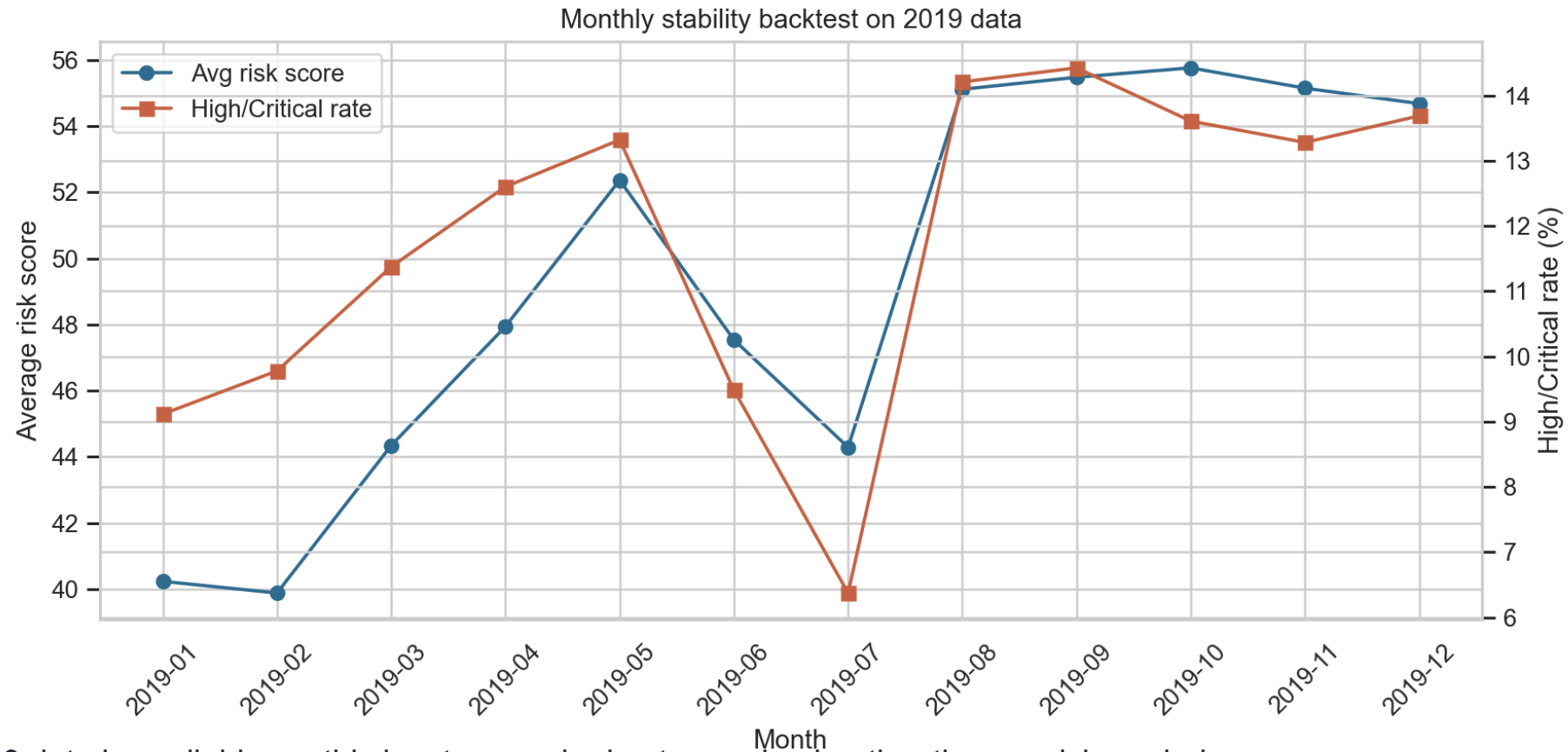
Key Insights

Top-risk queue is explainable by root-cause branch



Temporal Stability

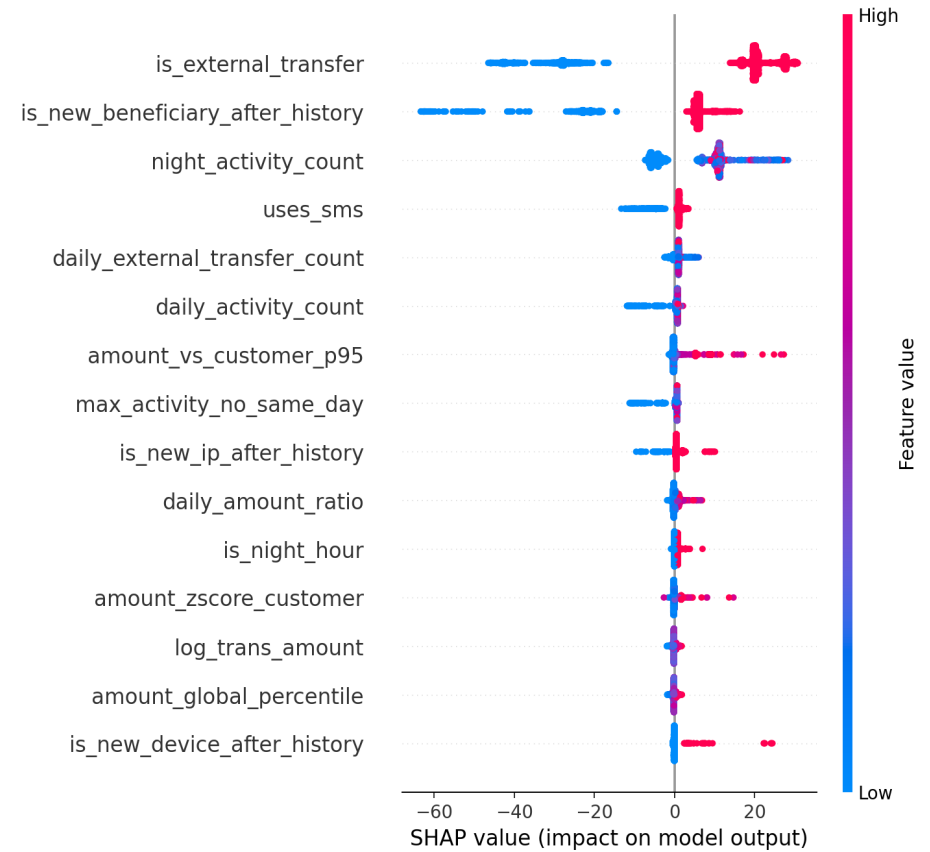
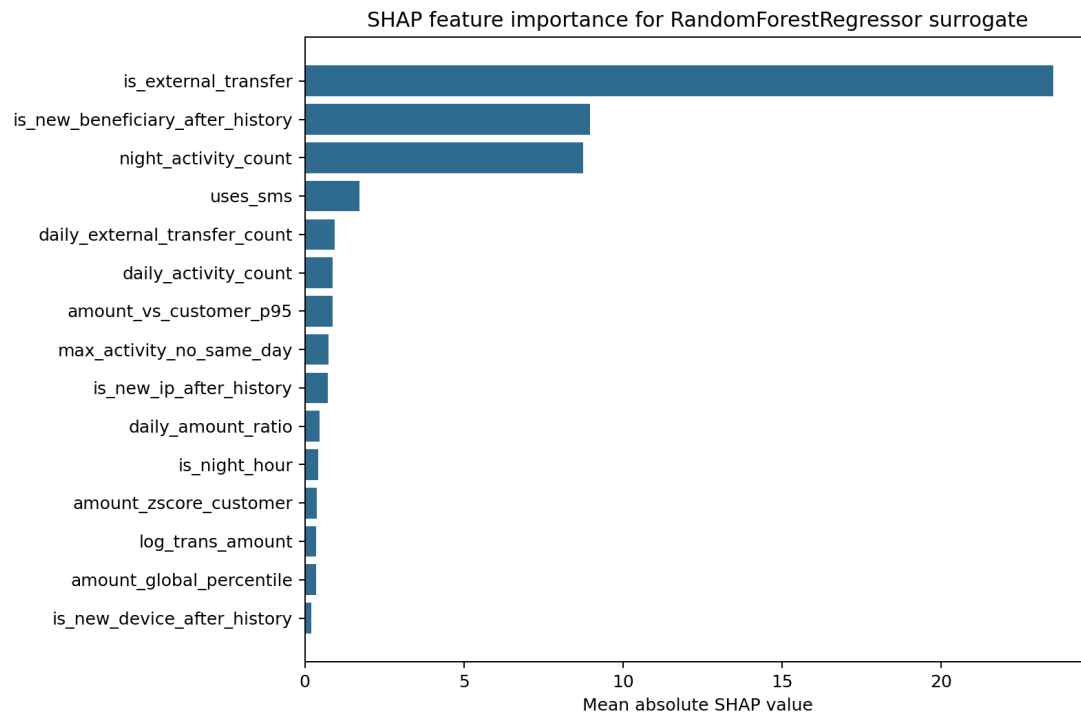
2019 monthly/quarterly backtest for review-rate stability



- Only 2019 data is available, so this is a temporal robustness check rather than a crisis-period validation.
- The KPI is not confirmed-fraud accuracy; it is weak-label prevention coverage, protected amount, and reason-code consistency over time.

xAI Engine

SHAP explains a tree surrogate of the supervised prevention score



Operational Recommendations

Turn model output into bank actions

- Critical: temporary hold, near-real-time manual review, customer verification, step-up authentication.
- High: step-up authentication before allowing the transaction to proceed.
- AML branch: network escalation rather than single-transaction review only.
- Medium: enhanced monitoring; escalate if repeated within the next review window.
- Feedback loop: investigator decisions become labels for Precision@K, Recall@K and supervised model tuning.

Live Demo

Streamlit advisor for transaction/customer review

- Run ``streamlit run src/demo_app.py``.
- Select a top-risk transaction or enter a CUSTOMER_NUMBER.
- The advisor returns risk band, root-cause branch, reason codes, recommended action and SHAP evidence.
- This is designed for a short judging demo or screen-recorded walkthrough.